

**PRAGATI ENGINEERING COLLEGE: SURAMPALEM
(AUTONOMOUS)**

II B.Tech I Semester Regular Examinations, November-2024

**OBJECT ORIENTED PROGRAMMING THROUGH JAVA
(Common to CSE, IT, CSE(AI&ML),CSE(AI),CSE(DS) and CSE(CS))**

Time: 3 hours

Max. Marks: 70

Note:

- i. Question No. 1 shall contain 10 compulsory short answer questions (2 questions from each unit) for a total of 20 marks such that each question carries 2 marks.
- ii. In each of the questions from 2 to the last question, there shall be either/or type questions of 10 marks each.

Q. No.	Question	BTL	CO	Marks
1.	a) Define abstraction and polymorphism.	K1	CO1	2M
	b) What are the command line arguments?	K2	CO1	2M
	c) Define a class and object. Give the syntax for creating an object.	K1	CO2	2M
	d) What happens when you declare a method as 'final'? give the syntax.	K1	CO2	2M
	e) What is an array? How do you declare and initialize a single dimensional array?	K2	CO3	2M
	f) What is the purpose of super keyword?	K1	CO3	2M
	g) List any four built-in packages from Java API along with their use.	K2	CO4	2M
	h) List any two byte and character stream classes.	K1	CO4	2M
	i) What is the difference between a String object and StringBuffer object?	K2	CO5	2M
	j) Define an event. List the packages to handle events.	K1	CO5	2M
UNIT-I				
2.	a) Write short note on different data types in Java.	K1	CO1	5M
	b) How do you comment in Java program. Write example program for single line and multi-line comments.	K2	CO1	5M
OR				
3.	a) What is the advantage of for-each loop? Write a Java program using for-each loop to print the sum of elements of an array.	K3	CO1	5M
	b) Write short notes bitwise logical operators.	K1	CO1	5M

UNIT-II					
4.	a)	Write a program in Java to show how to pass arguments by reference.	K2	CO2	5M
	b)	What is a nested class? How do you access the members of a nested class? Give example.	K2	CO2	5M
OR					
5.	a)	Explain the concept of method overloading with an example program.	K2	CO2	5M
	b)	Why is main declared as public static and void? Explain.	K2	CO2	5M
UNIT-III					
6.	a)	What are the different operations possible on the elements of an array? Give suitable examples.	K2	CO3	5M
	b)	Explain the concept of single inheritance with suitable programs.	K2	CO3	5M
OR					
7.	a)	What is importance of super and this keyword in inheritance? Illustrate with suitable example.	K2	CO3	5M
	b)	Can a class implement multiple interfaces? If so give the syntax and explain with examples.	K3	CO3	5M
UNIT-IV					
8.	a)	What are the key words used to handle exceptions in java? Explain with an example.	K2	CO4	5M
	b)	Define a package. Explain how to create used defined packages.	K2	CO4	5M
OR					
9.	a)	Explain how wrapper class objects are automatically converted back into primitive types.	K2	CO4	5M
	b)	Write a program to read a file using character stream.	K3	CO4	5M
UNIT-V					
10.	a)	Explain the life cycle of a thread.	K2	CO5	5M
	b)	Write a program to handle mouse events.	K3	CO5	5M
OR					
11.	a)	Discuss how to create thread using Runnable interface.	K3	CO5	5M
	b)	Explain JDBC architecture.	K2	CO5	5M